

Steven Jonathan Sihombing

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SUMMARY

Informatics student at Diponegoro University specializing in full-stack JavaScript/TypeScript development and microservices architecture. Adept at designing scalable RESTful APIs, event-driven systems, and data pipelines using Node.js and Golang. Experienced in working within Agile environments and utilizing CI/CD pipelines to deliver high-quality, test-driven applications. Eager to apply best practices in OOP and SOLID principles to build robust enterprise solutions.

EDUCATION

Diponegoro University	Semarang, Indonesia
<i>Bachelor of Computer Science (Informatics) / GPA: 3.70</i>	<i>2023 - present</i>

WORK EXPERIENCES

Telkom Regional 4	Semarang, Indonesia
<i>Full Stack Developer Intern</i>	<i>January 2026 - February 2026</i>

- Architected and deployed a production-ready FTTH monitoring dashboard from scratch, streamlining project tracking for field divisions. Implemented defensive UX design and robust error handling to mitigate slow client networks, which significantly reduced user complaints and simplified tasks for non-technical workers.
- Secured internal corporate data access by engineering a Telegram bot integration featuring a strict, database-driven phone number authentication system, successfully ensuring 100% authorized usage.

DBS Foundation Coding Camp	Remote
<i>Full-Stack Web Developer</i>	<i>February 2026 - July 2026</i>

- Selected as 1 of only 3,000 competitive participants from 67,000+ national applicants (~4.4% acceptance rate) to undergo an intensive, high-standard digital talent development program.
- Invested 943+ hours mastering comprehensive software engineering, covering modern frontend (React, Vite, Zustand), scalable backend (Node.js, Express, RESTful APIs), database management (PostgreSQL/Supabase), Redis caching, and AWS cloud.
- Collaborated in a cross-functional team to co-develop 'InvestSense AI', leveraging SOLID principles to architect REST-based APIs and microservices. Implemented background processing and performance tuning via Redis caching to ensure optimal system reliability.

PROJECTS

WalletX (Personal Finance & Budgeting Assistant)	
<i>Software Engineer</i>	<i>August 2026 – Present</i>
Tech Stack: Golang, PostgreSQL (Supabase), Redis, Vector Database, Gemini AI, Telegram Bot API, Docker.	

- Built a production-ready backend in Golang using Layered Architecture (Handler-Service-Repository) and Test-Driven Development, with tests for concurrent operations and race conditions.
- Improved database performance and data safety by using type-safe SQL, transaction locking, and Common Table Expressions (CTEs) to maintain data consistency during concurrent requests.

- Built a secure AI-powered RAG pipeline using Gemini AI for automatic transaction extraction and semantic search, with strict per-user data isolation.
- Improved API security and data integrity using DTOs and custom validation to prevent mass assignment and hide sensitive database errors from API responses.

InvestSense (AI-Powered Investment Assistant)

Backend Developer

March 2026 - July 2026

Tech Stack: Node.js (Express), Python (FastAPI), PostgreSQL (Supabase), Redis, Yahoo Finance API, GNews API, Vercel, Hugging Face Spaces.

- Architected scalable RESTful APIs in Node.js using Clean Architecture, integrating a multi-tier Redis cache with namespaced keys for stock quotes, RSI indicators, and RAG chatbot context, with fail-open error handling ensuring graceful degradation on cache outages.
- Built an automated daily data pipeline orchestrating Yahoo Finance OHLCV ingestion, RSI-14 calculation via a Python microservice, and AI-powered news sentiment analysis for watchlisted IDX stocks, with Supabase upsert idempotency to eliminate duplicate records.

Diponegoro Chemistry Fair (DCF) 2026

Project Manager / Lead Developer

March 2026 - September 2026

Tech Stack: Next.js, Supabase, PostgreSQL, Docker, Nginx, Moodle, k6.

- Engineered a high-performance participant registration portal using Next.js and Supabase, integrating Framer Motion for premium UI micro-interactions and Nodemailer for automated email workflows.
- Architected a scalable Computer-Based Test (CBT) infrastructure by containerizing Moodle, Nginx, and PostgreSQL via Docker, and validated system resilience using k6 automated load testing to guarantee zero downtime for 250 concurrent users during peak exam periods.

SKILLS

Tools/Aplikasi: Git, GitHub, Docker, Figma, VS Code, Postman, Vercel.

Hard Skills/Teknikal: React.js, Next.js, TypeScript, Golang, Flutter, Node.js, PostgreSQL (Supabase), Redis, Tailwind CSS, RESTful API Design, Clean Architecture, SOLID Principles, Responsive Web Design.

Soft Skills/Teknikal: Problem Solving, Cross-functional Communication, Stakeholder Collaboration, Agile/Scrum, Analytical Thinking.

Language/Bahasa: Indonesian (Native), English (Professional Working Proficiency).
